Aadithyaa R

aadiramki01@gmail.com | GitHub | LinkedIn | Portfolio

Education

Shiv Nadar University Chennai, India
B.Tech Computer Science (IoT) || CGPA – 8.55 *Aug 2023 – May 2027*

Experience

Hypertangent Technology Chennai, India
Machine Learning Intern *June 2026 – Present*

- Engineered async ROS 2 vision preprocessor synchronizing color and depth feeds from Intel RealSense, generating dense observation tensors.
- Developed PPO-based RL policies using Stable-Baselines3 and Gymnasium for autonomous rover navigation and path planning.
- Integrated dynamic reward shaping in Gazebo Ignition to optimize mobile robot control algorithms.

Projects

DistroStore – Distributed File Storage System | Go, Consistent Hashing, Raft, SHA-256 GitHub

- S3-inspired distributed storage: streaming chunked uploads, multi-node replication, and fault-tolerant reads with replica fallback – file never fully buffered in gateway memory.
- Content-addressable chunks via SHA-256; benchmarked hash ring with 150 vnodes/node – **9.34% max load imbalance** and **22.8% key migration** on node addition over 100k keys.
- Chunk rebalancer migrates only ownership-changed chunks on node join; metadata persisted as atomic JSON snapshots – gateway restarts recover full file state without data loss.
- Raft-style leader election (randomized timeouts, majority votes, heartbeats) with transparent non-leader proxy; optional constant-time API-key auth on both gateway and storage layers.
- Exponential backoff on write (200/400/800ms) and read paths; 27 tests across 5 packages including mock HTTP server tests verifying actual backoff timing.
- Designed a pluggable storage-node interface decoupling the gateway from the underlying disk layer, enabling in-memory nodes for testing and local-disk nodes for production without code changes.
- Exposed a lightweight gossip-based membership protocol for node discovery and failure detection, keeping the hash ring consistent across the cluster without a centralized coordinator.

QEMULAB – Virtual Network Topology Builder | React, Node.js, Docker, QEMU, Guacamole GitHub

- Browser-based virtualization lab orchestrating **KVM-accelerated QEMU** instances to simulate complex network topologies with real-time device interaction.
- Integrated **Apache Guacamole** with custom **PostgreSQL** backend for low-latency **VNC/Telnet** sessions in-browser – no client install required.
- Custom networking layer using **multicast sockets** to simulate Layer 2 connectivity with dynamically isolated network segments.
- Built a drag-and-drop React canvas for composing topologies (routers, switches, hosts) that compiles to Docker Compose and QEMU launch configs, removing manual VM setup.
- Containerized the orchestration layer with Docker for reproducible lab environments; automated teardown and resource cleanup to prevent orphaned QEMU/Guacamole sessions.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, Golang
Frameworks: React.js, Next.js, Node.js, Express.js, Gin
Tools: Git, Docker, QEMU, Firebase, ROS 2, Gazebo Ignition
ML/AI: Stable-Baselines3, Gymnasium, Google Gemini
Databases: MongoDB, PostgreSQL, Redis, Firestore

Certifications

- DBMS - SQL AND NoSQL
- DBMS - with JDBC
- Data Structures Using Java
- Programming Using Java
- Agile Scrum in Practice
- Software Engineering and Agile